



Research on nozzle and needle combination for high frequency piezostack-driven dispenser

Xiang Huang^a, Siying Lin^a, Zhenxiang Bu^a, Xiaolong Lin^a, Weijin Yi^a, Zhihong Lin^a,
Peiqin Xie^a, Lingyun Wang^{a,b,*}

^a Xiamen University, Department of Mechanical and Electrical Engineering, Xiamen, 361024, China

^b Shenzhen Research Institute of Xiamen University, Shenzhen, 518057, China

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ABSTRACT

An on-off valve piezostack-driven dispenser is used as a study object to investigate the influence of the needle and nozzle combination on the high frequency jetting. Firstly, the jetting dispensing process and fluid dynamics are studied. Secondly, FLOW-3D software is used to model and simulate the needle and nozzle with different mating angles of sphere-sphere, sphere-short cone, sphere-short cone and cone-cone. The operating performances of different matching forms are compared and summarized according to the simulation results. Thirdly, the designed spherical-spherical needle and nozzle with 180° are grinded and installed on the dispenser, and the experiment system is built. By using a glycerol/alcohol mixture with a viscosity of 180 cP, we can realize the industrial frequency of 360 Hz in droplet continuous dispensing, and the average diameter of droplets is 525 μm. Moreover, the viscosity of 800 cP industrial epoxy resin form droplets have good geometry. under the conditions of nozzle diameter of 100 μm and square-wave frequency of 320 Hz, the variation of the droplet diameter is within ±2%. The findings in this paper can guide the design of nozzle and needle for jetting dispenser, which is used to meet specific industrial requirements.

1. Introduction

Fluid dispensing technology distributes certain viscosity fluids in a controlled manner in order to realize products adhesion and fixation. With the development of integration and miniaturization of electronics [1], high speed and accurate micro-distribution of adhesives is necessary in the microelectronics domain. According to the relative position between the nozzle and the substrate, it can be categorized as contact dispensing and jetting dispensing [2,3]. The disadvantages of traditional contact dispensing, such as low efficiency, great workspace, large size and poor consistency, has been unable to meet the production demands, and is gradually replaced by jetting dispensing [4–6]. Jetting dispensing technology [7–9] separates the fluid by applying pressure in the nozzle, forming the droplets on the substrate. In the process of product packaging, it can reduce the contact between the needle and products and effectively avoid the disadvantages such as drawings and scratches. Besides, the dispenser has no vertical displacement, which greatly improves the distribution efficiency and the uniformity of droplets.

Piezostack-driven dispensing technology, as one of non-contact

dispensing technology, has many insightful advantages, such as adhesives are widely applicable, droplets are easy to control and working efficiency is high. Therefore, many scientific institutions have pursued extensive research. Sun et al. [10,11] presented a new type of piezostack-driven jetting dispenser, which can be applied to dispense high viscosity adhesives in microelectronic packaging. It was able to allocate adhesives whose viscosity is less than 7000 cP and obtain a minimum droplet volume of 0.055 μL. But when the operating frequency reaches 120 Hz, the output displacement of the needle will rapidly decrease. Deng et al. [12,13] designed a direct-acting piezoelectric jet dispenser with rhombic mechanical amplifier. It could jet 1400 cP adhesive without preheating and deliver 100 adhesive dots per second. The lifespan of the nozzle and needle exceeds 20 million cycles. Li et al. [14] proposed a silicone jet dispensing for light-emitting diode packaging and studied the breakup behavior of the jet dispensing process. The results indicate that the nozzle outlet is at least 2 mm from the substrate is beneficial for jet breakup and high-volume consistency, also the disturbance wavelength between 0.3 and 0.6 is beneficial for both thread breakup and high-volume consistency. Nguyen et al. [15,16]

* Corresponding author. Xiamen University, Department of Mechanical and Electrical Engineering, Xiamen, 361024, China.

E-mail address: wangly@xmu.edu.cn (L. Wang).

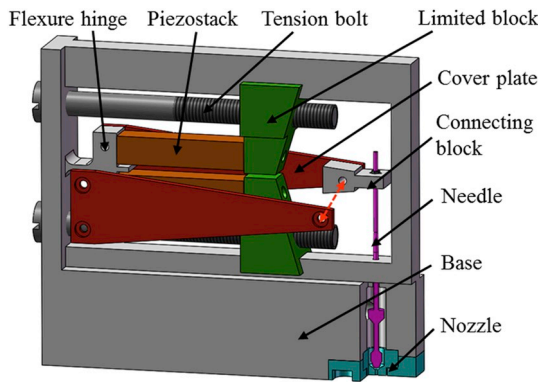


Fig. 1. Structure of piezostack-driven dispenser.

proposed a new type of jetting dispenser driven by a piezostack and a flexible beam. It could provide a very small dispensing dot size of a low viscous adhesive, from 50 to 500 cP at a high dispensing flow-rate in semiconductor packaging processes. Choi et al. [17,18] presented a double piezoelectric ceramic dispenser based on lever principle, and studied the effects of pre-tightening force, square-wave frequency, pulse time, nozzle diameter and temperature on the formation of the droplets. Its maximum operating frequency higher than 300 Hz, and the response time of less than 0.35 ms. Furthermore, various commercial piezoelectric dispensing systems are widely used in industry, which have excellent performance in microdroplet dispensing, high viscosity adhesives distribution and applied fluid diversity [19–22]. Generally, these dispensing systems are capable of jetting various fluids with different material composition, which have a wide range of fluid viscosity applications. But when applied fluid is determined, it takes a large amount of time for dispensing tests. By changing the parameters of nozzle opening time, needle stroke and speed to obtain better operating performance, but it reduces the production efficiency, easily causes damage to the piezostack in heavy load, and has high experience demands for operators. Thus, the research on nozzle and needle combination is needed to guide the design of high performance jetting dispenser, which is used to meet specific industrial requirements.

Previously, we mainly focused on the design of piezostack amplification mechanism and developed a double piezoelectric dispenser with triangular magnification mechanism [23]. Its maximum jetting frequency is 65 Hz, and droplets of 1.07 mm diameter were produced by a stainless steel nozzle of 0.25 mm diameter in the experimental. Although the operating frequency of dispenser is small, it has accumulated valuable experience for our subsequent research. This study mainly focuses on the combination of the nozzle and needle on the different operating performances of the dispenser. On the basis analysis of the jetting dispensing process and fluid dynamics, the FLOW-3D software is used to model and simulate the needle and nozzle of different matching forms. Then, the parameters of nozzle chambers were compared, and the performances of nozzle chambers are summarized. Also, some strategies are given to improve the performance of the dispenser. Finally, the spherical-spherical needle and nozzle with the matching angle of 180° is processed and installed on an on-off valve piezostack-driven dispenser to test the high-frequency jetting performance and consistency of droplets.

2. Dispensing process and jetting principle

2.1. Jetting dispenser process

For the purpose of studying the influence of the needle and nozzle matching form on the high performance jetting, the structure of jetting dispenser used in this paper is shown in Fig. 1. Two piezostacks are symmetrically fixed on the flexible hinge arm, and the pre-tightening force is provided by a tension bolt adjusted limit block. During

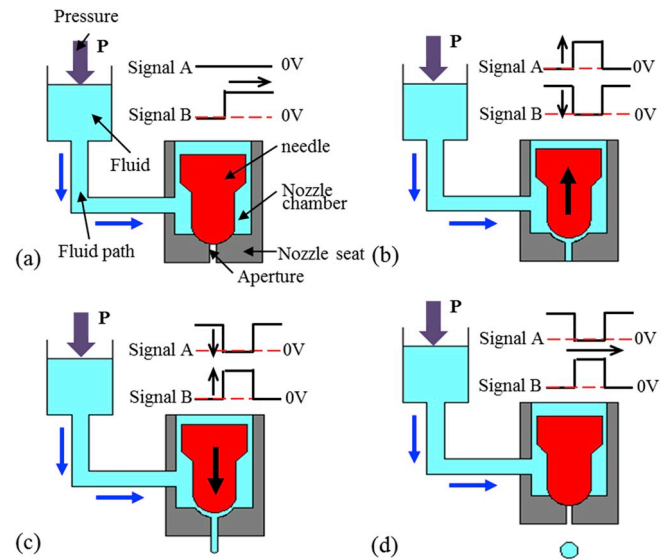


Fig. 2. Nozzle stage during dispensing: (a) dispenser power-on (b) needle rising (c) needle falling (d) nozzle shut.

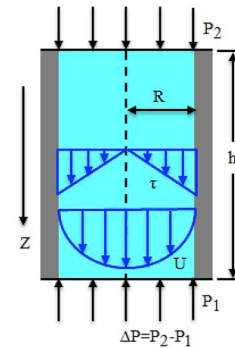


Fig. 3. Dynamics model of fluid in the nozzle.

dispensing, the upper and lower piezostacks elongate alternately, which drives the end of displacement amplifier to rotate at a small angle. Meanwhile, the material cylinder is required to be continuously supplied fluids for dispenser. Under the effect of external pressure, the fluid is always filled into the nozzle chamber through the runner. Fig. 2 shows the working state of the nozzle, where signals A and B respectively represent the signals connected by the upper and lower piezostacks. As shown in Fig. 2a, when the dispenser is energized, the lower piezostack extends due to the direct current signal, which makes the needle and the nozzle seat are tightly fitted. From Fig. 2b and c, signals A and B alternately generate square wave pulses, which control the elongation and restoration of the upper and lower piezostacks respectively. It causes the needle to rise and fall, squeezing the fluid out of the nozzle. Fig. 2d shows that when the needle contacts the seat again, the fluid stops flowing. Due to the shear force produced by the impact and the surface tension of the fluid, the jet breaks and gradually forms a rounded droplet.

2.2. Fluid dynamics analysis

Usually, the type of jet flow in dispenser is laminar and incompressible, which is consistent with the power-law fluid model [3,24]. By ignoring the slip effect and shear force on fluid viscosity, considering the fluid material properties are continuous, the dynamics model of fluid in the nozzle is shown in Fig. 3. Gravity of fluid is negligible, and the dynamics equation can be expressed in cylindrical coordinates as:

$$\rho \frac{dU}{dt} - \frac{1}{r} \frac{\partial}{\partial r} (\tau r) = - \frac{dP}{dz}, 0 \leq r \leq R \quad (1)$$

where ρ is the density of the fluid, U is the velocity of the flow, τ is the shear stress along the radius r . By assuming that the pressure is linearly distributed on the axis, the above equation can be converted to:

$$\rho \frac{dU}{dt} - \frac{1}{r} \frac{\partial}{\partial r} (\tau r) = \frac{\Delta P}{h}, 0 \leq r \leq R \quad (2)$$

where h is length of nozzle, ΔP is the pressure difference between the inlet and outlet of the nozzle. In the power-law fluid model, there is:

$$\tau = -K \left(-\frac{\partial U}{\partial r} \right)^n \quad (3)$$

where, K is the dynamic viscosity (Pa·s), and n is the flow coefficient. When the fluid flow type is laminar flow, $n=1$. Moreover, in steady flow, $\frac{dU}{dt} = 0$. Therefore, equation (2) becomes:

$$\frac{K}{r} \frac{\partial}{\partial r} \left(\left(-\frac{\partial U}{\partial r} \right)^n r \right) = \frac{\Delta P}{h}, 0 \leq r \leq R \quad (4)$$

Under no-slip boundary conditions, $U(r=R) = 0$ and $\tau(r=0) = 0$. We can derive that:

$$U(r) = \frac{n}{n+1} \left(\frac{\Delta P}{2hK} \right)^{\frac{1}{n}} \left(R^{\frac{n+1}{n}} - r^{\frac{n+1}{n}} \right) = \frac{\Delta P}{4hK} \left(R^2 - r^2 \right), (n=1) \quad (5)$$

The flow volume has a mathematics relationship with the flow rate, which is:

$$Q = \int_0^R 2\pi r U(r) dr = \frac{\pi \Delta P}{8hK} R^4 \quad (6)$$

According to equation (6), the average flow velocity $\bar{U}(R)$ and jet volume $V(R)$ can be obtained as:

$$\bar{U}(R) = \frac{Q}{\pi R^2} = \frac{\Delta P}{8hK} R^2 \quad (7)$$

$$V(R) = \int_0^t Q dt = \frac{\pi \Delta P}{8hK} R^4 t \quad (8)$$

Equations (7) and (8) show that when the nozzle radius increases, the jet velocity increases by a factor of square, but the jet volume increases by a factor of four-order. Therefore, this scheme of achieving high frequency dispensing is generally not adopted. Moreover, the length of the nozzle cannot be too small, which may affect the verticality of the jet due to the eccentricity of the needle and nozzle assembly. The method of controlling the velocity of jet flow by changing the pressure inside nozzle is effective.

Moreover, overcoming viscous forces and surface tension is a prerequisite for fluid dispensing, which is related to the dimensionless number Weber number W_e and Reynolds number Re , respectively [25]. They need to meet the following conditions:

$$W_e = \frac{\rho U^2 d}{\sigma} > 1 \quad (9)$$

$$Re = \frac{\rho U d}{\mu} > 1 \quad (10)$$

where, σ is surface tension coefficient, μ is hydrodynamic viscosity, and d is nozzle aperture diameter. When the velocity meets the conditions of inequalities (9) and (10), the jet flow can break and fall onto the substrate smoothly.

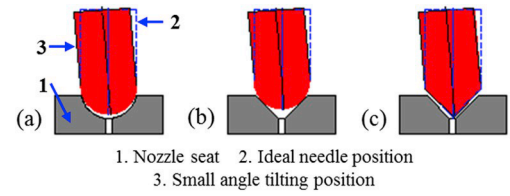


Fig. 4. Various nozzle match types: (a) spherical-spherical (b) spherical-conical (c) conical-conical.

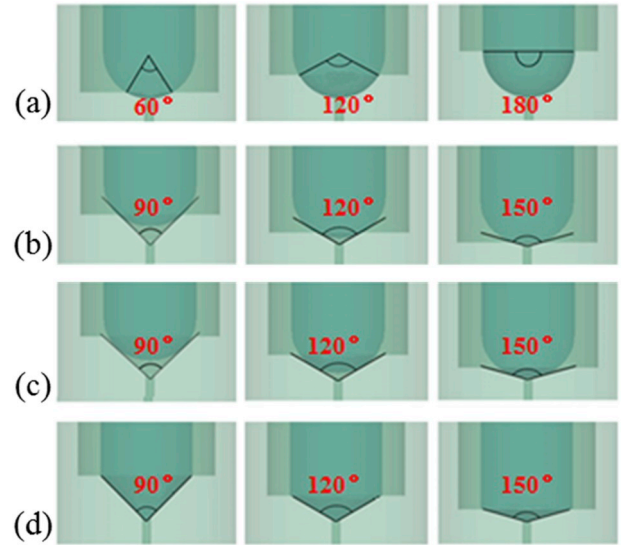


Fig. 5. Different types of nozzle cavity fit model: (a) Type I (b) Type II (c) Type III (d) Type IV.

3. Combination of nozzle and needle

The study on the combination of nozzle and needle is important because it can effectively prevent fluid leakage with the effect of closing when the dispenser in the normal shutdown [5]. Under the same drive, different nozzle and needle matching form has great influence on the parameters of the jet. As part of the nozzle, the nozzle chamber consists of a needle and a seat. It plays the role of storing fluids and providing jetting pressure in dispensing process. In general, the on-off valve needle and seat are often machined to spherical or conical surfaces, and there are three possible needle-seat matching forms (Fig. 4), spherical-spherical match, conical-conical match and spherical-conical match. Fig. 4a shows the spherical-spherical combination, which has the advantages of good tightness and little influence of small angle inclination of needle, but it is difficult to machining. Fig. 4b shows the spherical-conical combination, which has poor tightness but less affected by small angle tilting of needle and the needle is a little difficulty in machining. Fig. 4c shows the conical-conical combination, which has the advantages of good tightness and easy processing, but the small angle tilting of the needle has a great influence.

3.1. Nozzle chamber modeling and simulation

FLOW-3D simulation software is used to simulate the needle and seat with different matching forms. During the simulation, the uniform parameters of nozzle chambers with different mating modes were set as follows: nozzle aperture 0.2 mm, nozzle length 0.6 mm, nozzle chamber internal diameter 3 mm and needle diameter 2 mm. According to the variation range of nozzle parameters commonly used in practice, the different nozzle models used in the simulation are established as shown in Fig. 5. It respectively shows the spherical-spherical, spherical-short

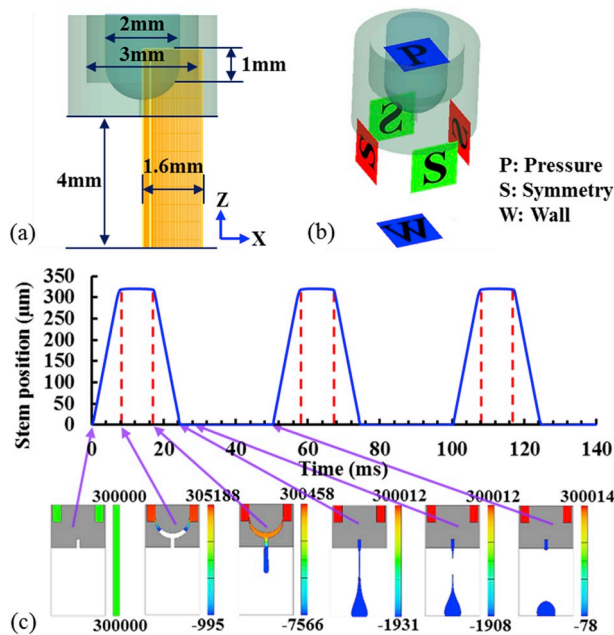


Fig. 6. Simulation parameters and droplet formation process: (a) mesh generation (b) boundary condition (c) needle displacement curve and droplet formation process.

conical, spherical-long conical and conical-conical fit models with different matching angles. Moreover, in order to facilitate the subsequent analysis of the simulation results, four nozzle matching forms are referred to as type I-IV.

As the nozzle and the jet are all axisymmetric around the center, the mesh is simplified to two dimensions in XZ panel to reduce the computational space (see Fig. 6a). The mesh is distributed on one side of the central axis of the nozzle, and the width (X direction) is 1.6 mm, which is divided into 55 equal segments. The height (Z direction) is 6.5 mm, and is divided into 220 equal segments. The upper boundary of the grid is 1 mm higher than the contact point between the needle and the seat, the lower boundary is 4 mm away from the nozzle outlet and the outer boundary is 0.1 mm away from the nozzle chamber. Grid boundary conditions are shown in Fig. 6b. Due to the fluid needs

continuous pressure supply, the upper boundary of the grid is set as the constant pressure condition, and the feeding pressure is 3 bar. The lower boundary is wall, which is used to simulate droplet collection. In the simulation, the fluid will not touch the outer boundary of the grid, and the boundary conditions can be set to be symmetrical. In general, the fluid is laminar flow in the nozzle chamber, and the grid boundary near the central axis can be set as a symmetrical boundary condition, on which the fluid does not have transverse convection. Besides, glycerol was selected as the jet fluid in the simulation. The density of the glycerol is 1263.31 kg/m^3 , the dynamic viscosity is 1412 cP and the surface tension coefficient is 0.0633 N/m at 20°C . The needle in the dispenser nozzle has a movement relative to the seat, setting the needle displacement curve shown in Fig. 6c. The maximum displacement of the needle is $320 \mu\text{m}$, the period is 50 ms , the duty cycle is 35% and the maximum speed for needle rising and falling is 44 mm/s .

The droplet formation process in one cycle is also shown in Fig. 6c. When the needle rapidly rises from the lowest position, a crescent-shaped cavity is formed between the nozzle seat and the needle, and a small amount of fluid enters the cavity from the gap. When the needle reaches the highest position and maintains the lifted state, the fluid fills the crescent-shaped cavity under the action of back pressure and gravity, and flows out from the nozzle outlet. When the needle reaches the highest position and remains lifted, the fluid fills the crescent-shaped cavity under the action of gravity and feed pressure, then flow out of the nozzle outlet. After, the needle falls quickly and the fluid is forced out of the nozzle to form a jet flow. With the needle back to the lowest position, the nozzle closed and the fluid stop flowing out. Due to the different velocity distribution of the fluid in the longitudinal direction, the jet tail is elongated and breaks under the action of surface tension. Eventually, the fractured fluid tail shrinks to form a rounded droplet [26].

3.2. Analysis of simulation results

When the dispenser works, it is required to distribute a variety of different fluids. However, the viscosity and surface tension coefficients of different fluids are quite different, and the lower the viscosity and surface tension coefficients, the easier to jetting. In order to achieve high viscosity or high surface tension coefficient of the fluid jetting, the nozzle must have a large jet velocity. The dispenser should have high frequency jetting performance to improve the efficiency of electronic

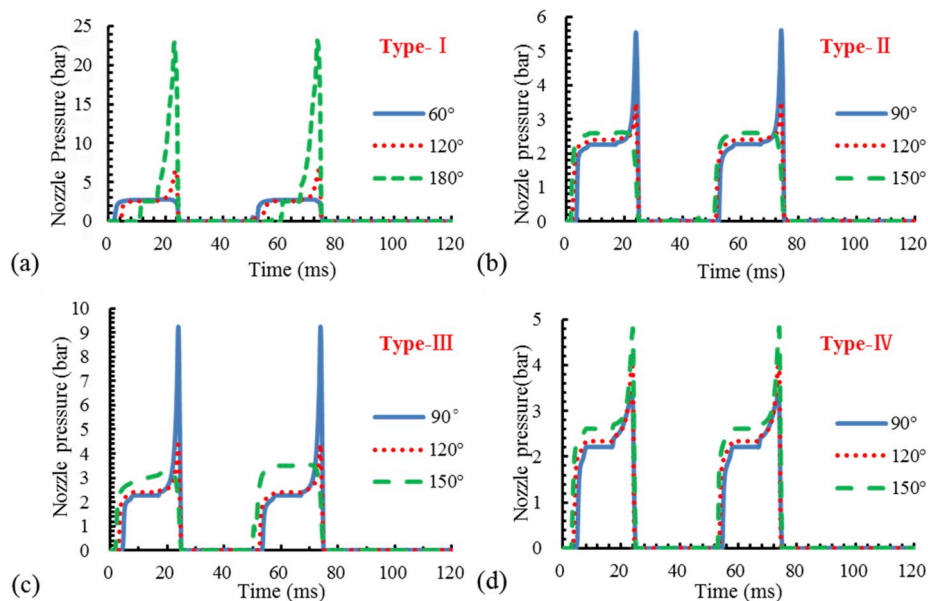


Fig. 7. Pressure variation of nozzle inlet with different mating modes: (a) Type I (b) Type II (c) Type III (d) Type IV.

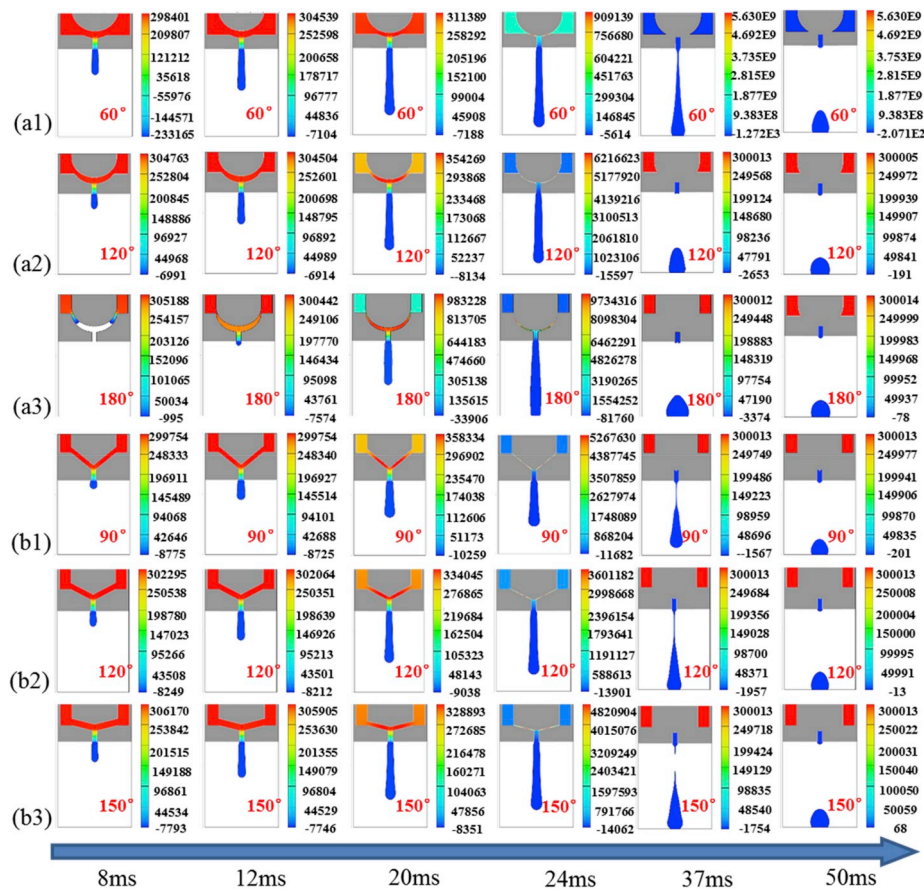


Fig. 8. Nozzle chambers pressure distribution: (a1)~(a3) Type I nozzle pressure distribution with contact angles of 60°, 120° and 180° (b1)~(b3) Type IV nozzle pressure distribution with contact angles of 60°, 120° and 150°.

packaging. Therefore, the time required for fluid flow from the nozzle chamber is short, otherwise, the fluid supply will not be timely. Moreover, when the fluid arrives at the substrate, a uniform round droplet is demanded. A faster relative movement of dispenser and packaging object will produce satellite drops or elliptical droplets [27]. Consequently, when the nozzle is closed, the jet flow must be broken as soon as possible.

It is the basic requirement of nozzle chamber design to be able to realize the jetting of different materials adhesive. Fig. 7 shows the variation trend of pressure with different nozzle chambers. As shown in Fig. 7a, the maximum pressure P of type I nozzle enlarge with the increase of nozzle and needle matching angle. When the matching angle is 180°, the maximum pressure is about 23 bar, which is obviously higher than other matching angles. From Fig. 8a1-a3, the pressure distribution of the type I nozzle chamber with different matching angles in a cycle. The larger matching angle makes the gap between the nozzle and the needle smaller, resulting it more difficult for fluid to flow into the cavity. However, the crescent cavity structure can effectively increase the reflux resistance and reduce the backflow of fluid, so that rapidly increase of the nozzle inlet pressure. For conical-conical nozzle chamber matching model with small contact angle, the nozzle inlet pressure has significantly increasing trend when the needle drops from the highest position. It is mainly due to the linear contact between the nozzle and needle, which causes a larger matching cavity with a smaller cone angle and squeezes more fluid in it. Compared with Fig. 7b and 7c, longer cone nozzle makes the nozzle and the needle create a smaller space near the contact line, which resists the backflow of the fluid. Therefore, the maximum nozzle inlet pressure of type III is larger than type II with matching angle of 90°. As for type IV (Fig. 7d), the nozzle inlet pressure increases with the enlarge of cone angle, and the sharp increase of

Table 1

Dispensing parameters with different matching forms.

Matching form	Type-I, 180°	Type-II, 90°	Type-III, 90°	Type-IV, 150°
Maximum fluid velocity	6.94 m/s	1.67 m/s	2.57 m/s	1.36 m/s
Average pressure	4.03 bar	2.12 bar	2.26 bar	2.26 bar
Outflow time	11.5 ms	4 ms	5.5 ms	4 ms
Reach substrate time	23.5 ms	24.5 ms	25.0 ms	30.1 ms
t_1				
Jet flow break time	31.5 ms	33.0 ms	37.5 ms	36.5 ms
t_2				
Jet duration time T	8.0 ms	8.5 ms	12.5 ms	6.4 ms

pressure can be obtained when the needle drops obviously. It is mainly due to the slope of the nozzle cone decreases with the increase of the matching angle (Fig. 8b1-b3), which facilitates the accumulation of the squeezed fluid toward the nozzle inlet. Furthermore, with the cone angle increases, the flow channel of the fluid along the cone becomes shorter and the pressure loss decreases. According to above analysis, the 180° type I nozzle, 90° type II and type III nozzles, 150° type IV nozzle have ability to achieve high nozzle inlet pressure in their same matching types respectively, which are easy to achieve dispense of fluid smoothly.

For further demonstrate the performance of the dispenser in different application requirements, we studied the jetting parameters with four different matching forms, as shown in Table 1. The maximum fluid velocity of nozzle outlet reflects the capability for overcoming the viscous forces and surface tension of fluid. Type I nozzle chamber with 180° matching angle makes the maximum fluid velocity up to 6.94 m/s, which has a wider range of applicable fluid viscosity compare with other

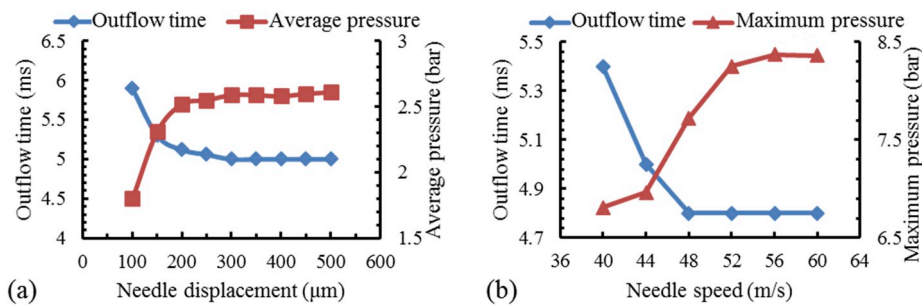


Fig. 9. Effects of needle displacement and speed on nozzle performance: (a) needle displacement (b) needle speed.

matching forms. In addition, the 90° type III nozzle chamber has maximum fluid velocity of 2.57 m/s, which is suitable for high viscosity fluid jetting when the processing conditions are restricted. The larger average pressure makes the fluid have greater kinetic energy, and usually squeezes more fluid out of nozzle in a dispensing cycle. Thus, the 90° type II nozzle chamber can easily realize microdroplet distribution. The outflow time of fluids from the nozzle chamber to nozzle outlet influences the high frequency jetting capacity of the dispenser. It is easy to realize high frequency when the outflow time is short. Otherwise, nozzle outlet cannot be timely supplied of fluid, and breaking the formation of droplets. So that, both 90° type II and 150° type IV nozzle chamber models have advantage of high frequency jetting. Since 180° type I matching form have small gap between needle and nozzle seat for fluid into, and the outflow time is long. The jet duration time T reflects the ability of dispenser to produce round droplets, which is the difference between the breaking time t_2 and reach the substrate time t_1 in a cycle, namely, $T = t_2 - t_1$. Among them, the 150° type IV matching model has a short jet duration time, which has greater chance to obtain good geometry of droplets. However, the jet flow reach substrate time and break time of type IV matching form are long, which limits the relative moving speed of dispenser and packaging products. Compare with it, the 180° Type I nozzle chamber has smaller reach substrate time and break time of jet flow, shows better performances in continuous lines and complex patterns formation.

In actual design of nozzle and needle, the specific industrial requirements need to be considered. Here are a few possible conditions: 1) For high viscosity fluid dispensing. The 180° Type I nozzle chamber model is recommended due to its excellent performance for realizing high fluid velocity of nozzle outlet, which supplies large shear force to conquer the viscous forces and surface tension of fluids. Nevertheless, high machining accuracy is required to ensure the sealing of the match. If the processing conditions limited, the 90° type III nozzle chamber is available. We can increase the feed pressure appropriately to suppress fluid reflux and design the side cage for needle to strength the shear force. 2) For microdroplet distribution. The 90° type II nozzle chamber is suitable and can properly reduce the feed pressure, nozzle diameter and

nozzle opening time. 3) For high frequency dispensing. The basic requirement is that outflow time is short to ensure timely fluid supply, so 90° type II and 150° type IV nozzle chamber can be chosen. In addition, increase the needle displacement and speed are useful for shorting the outflow time. But, large needle speed and displacement are not conducive to low viscosity fluids, which may cause energy imparted into the jet flow and form satellite droplet. 4) For good geometry of droplets. There are great requirements for the artistic process when packaging of the shell of some electronic products, and it needs good geometry of dispensing droplets. The 150° type IV nozzle chamber shows excellent performance due to short jet duration time. But, when the relative velocity of dispenser and products is fast, the 180° Type I nozzle chamber is better, which has short reach substrate time and break time of jet flow. Sometimes, we are supposed to consider the multiple operating properties of the dispenser. For 180° type I nozzle chamber model, it has a wide application range of fluid viscosity and good glue droplet geometry, but high frequency performance is limited by long fluid outflow time, which can be controlled by needle displacement and needle lift speed.

By exploring the effect of needle displacement and speed on jetting parameters, the performance advantages of type I nozzle chamber model is enhanced. The needle displacement can be controlled by the amplitude voltage of the square wave signal driving the piezostack. The effect of needle displacement on the nozzle performance parameters is shown in Fig. 9a. When the needle displacement increases from 100 μm to 300 μm, the fluid from the nozzle chamber outflow time t decreases gradually, and the pressure P at the nozzle inlet increases gradually. When the needle displacement decreases above 300 μm, there is no more decrease of outflow time t and the average pressure P . Besides, when the needle displacement increases, the opening time of the nozzle becomes longer in a single cycle, resulting in the enlargement of the droplet diameter. But in actual design, we should consider the influence of assembly clearance and other factors on the output displacement loss. Therefore, needle displacement should be slightly larger than 300 μm. The needle speed can be controlled by adjusting the time of rising and falling edge of square wave signals, and the influence on the

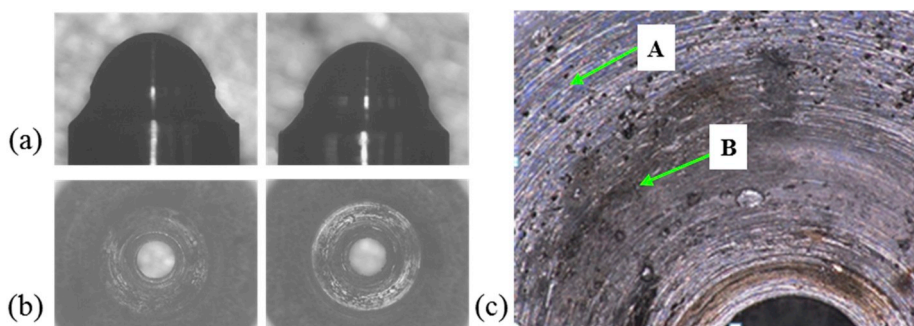


Fig. 10. Comparison between the needle and seat before and after grinding: (a) Needle head before and after grinding comparison (b) Seat internal before and after grinding comparison (c) Microscopic image of nozzle after grinding.

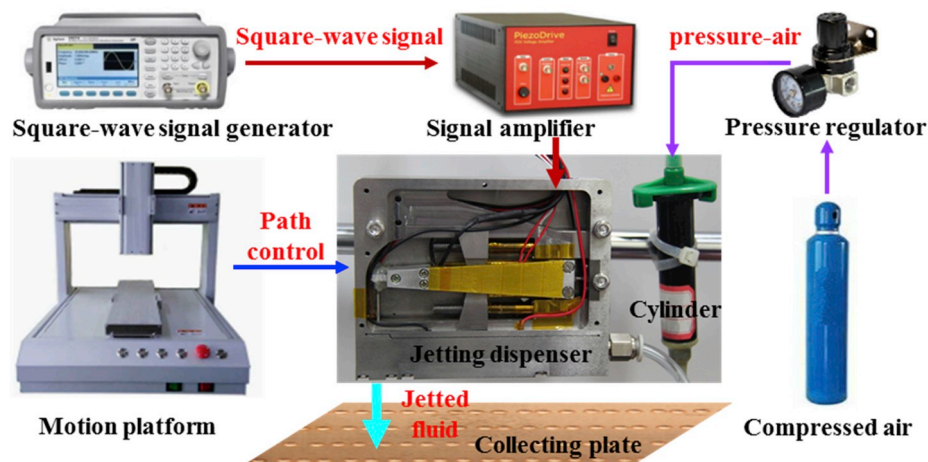


Fig. 11. Jetting dispensing system.

performance parameters of the nozzle is shown in Fig. 9b. Because the needle speed mainly affects the lifting and falling process of the needle, the outflow time and the maximum pressure of the nozzle cavity are significant factors. The outflow time decreases, while the maximum pressure increases of the needle speed from 40 m/s to 48 m/s. When the velocity is larger than 48 m/s, the maximum pressure continues to increase, but the outflow time has remained broadly stable. In order to improve the efficiency of piezoelectric dispenser, it is necessary to improve the response of piezostack, so the needle speed should be large.

3.3. Grinding of nozzle and needle

Type I nozzle chamber model is selected for test, but spherical-spherical combination of needle and seat machining is more difficult. Usually after the dispenser needle and nozzle are machined, further grinding is needed to enhance the sealing of nozzle chamber. In the grinding process, the abrasive in the cylinder enters the cavity between the needle and the nozzle through the runner, and the motor drives the valve needle to rotate and close to the seat for grinding until the grinding fluid no longer flows out of the nozzle. At this moment, the needle can be closely matched with the seat. Fig. 10a and Fig. 10b show the comparison between the needle and the seat before and after grinding. After lapping with a 120 nm abrasive solution, the hemispherical surface profile of the needle head is smoother, and the seat surface has less grain. After grinding, the nozzle was placed under the confocal laser scanning microscope for observation. As shown in Fig. 10c, the region B is rougher than region A, which indicates that the inner surface of the nozzle becomes smoother after grinding. For testing the grinding needle and nozzle on the dispenser and fill the cylinder with ethanol. When the feeding pressure is 7 bar and the nozzle is closed, the ethanol does not leak, which have achieved the working requirements of the dispenser.

4. Results and discussions of experiments

For testing the performance of developed needle and nozzle a dispensing system was installed. Fig. 11 shows a jetting dispensing test system consisting of dispenser and motion subsystem, drive subsystem and pneumatic subsystem. The motion subsystem controls the dispensing path by vision localization and program, the drive subsystem controls the status of the needle by generating square wave signals; and the pneumatic subsystem provides back pressure for the jetting fluid by adjusting the air pressure.

In the experiment, the high frequency jetting ability of dispenser is tested by adjusting the square-wave frequency under the conditions of glycerol/alcohol mixture viscosity of 180 cP, nozzle diameter 200 μm , square-wave amplitude voltage 180 V, duty ratio 50% and feeding pressure 6 bar. When the square-wave frequency increases from 240 Hz to 360 Hz, the change of droplet diameter is shown in Fig. 12a. As the square-wave frequency increases, the jetting period of dispenser decreases, the opening time of nozzle decreases, the volume of outflow fluid in one cycle decreases, and the droplet diameter decreases. Fig. 12b shows uniform circular droplets obtained with an average diameter of 525 μm at the frequency of 360 Hz. In addition, there is no leakage and media sticking at the nozzle during dispensing. Therefore, the dispenser has excellent high frequency performance.

In order to evaluate the uniformity of the jetting droplets, 20 droplets were randomly selected for measurement. The experimental conditions are as follows: nozzle diameter is 100 μm , adhesive viscosity is 800 cP, square-wave amplitude voltage is 180 V, frequency is 320 Hz and duty ratio is 40%. As shown in Fig. 13, the average diameter of the obtained droplets is 0.86 mm, the consistency deviation is $\pm 2\%$, and the droplets have good geometry.

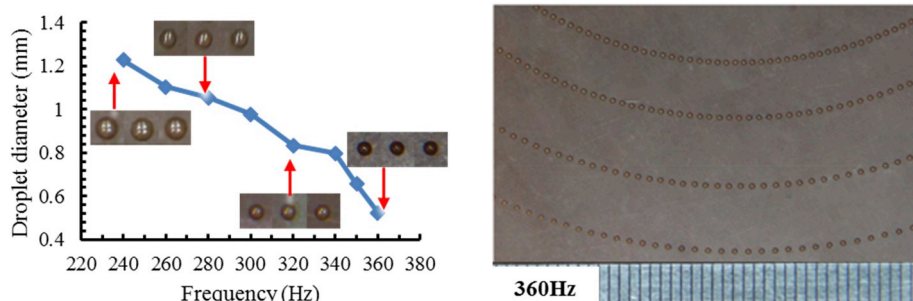


Fig. 12. The diameter of droplets at different frequencies.

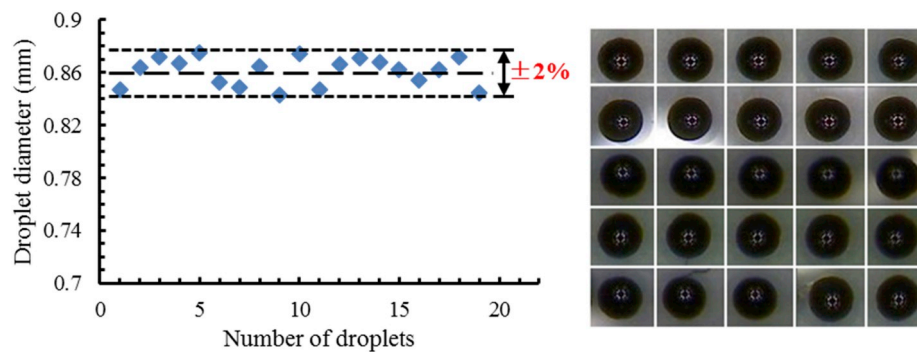


Fig. 13. Geometry of droplets.

5. Conclusion

In this paper, the influence of the combination of needle and nozzle on the performance of an on-off valve piezostack-driven dispenser is examined. On the basis of the jetting dispensing process study and fluid dynamics analysis, modeling and simulation of needle and nozzle in FLOW-3D software with the different angles of spherical-spherical, spherical-short conical, spherical-long conical and conical-conical are carried out. According to the simulation results, we have compared the performance of different nozzle and needle matching forms in high viscosity fluid dispensing, microdroplet distribution, high frequency operating and obtained droplets quality, which can be used to guide the design of jetting dispensing for some specific applications.

Furthermore, the 180° spherical-spherical needle and nozzle matching model is processed and used for testing, which has continuous operating frequency of 360 Hz. In high frequency application, the minimum droplet of 525 μm and the consistency of droplets within ±2%, and it can meet the basic needs of the electronic packaging industry. Of course, we have only verified the operating performance of one nozzle chamber matching form. For our subsequent work, we will test the performance of other matching forms in high viscosity fluid dispensing and microdroplet distribution. Meanwhile, we will pay attention to the optimization of displacement amplification mechanism and the design of piezostack driving power supply, and strive to achieve a higher dispensing frequency.

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